# Giovanni **Picogna**

Computational Physicist

*Munich, Germany*

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## Profile

Computational physicist with twelve years of experience writing, validating and maintaining multiphysics simulation codes. At LMU Munich I lead the radiation-hydrodynamics modelling of my group: from the governing equations and their discretisation, through solver coupling and parallelisation on national supercomputers, to public release. The resulting model grids are used directly as input by external research groups, so the code has to be correct, documented and stable in other people's hands.

Fast models for expensive physics: co-developed autoencoder surrogates that reduce a chemical reaction network 10× and halve total simulation time, checked against the full network on held-out cases. C/Fortran/Python, MPI/OpenMP, PyTorch; NVIDIA SciML/PINN (2022) and Deep Learning (2020) training.

## Technical Skills

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| **Simulation & HPC** | Radiation-hydrodynamics, magnetohydrodynamics (incl. Hall-MHD), finite-volume/finite-difference methods, coupled multiphysics solver design; MPI, OpenMP, SLURM; performance profiling and strong/weak scaling studies; Tier-0/Tier-1 HPC (SuperMUC-NG, ForHLR) |
| **Simulation Codes** | PLUTO (MHD), FARGO3D, PHANTOM (SPH), RADMC3D, MOCASSIN; own photoionisation/dust radiative-transfer and thermochemical solvers |
| **Programming** | C/C++, Fortran, Python (NumPy, SciPy, PyTorch, Astropy), Bash/Shell |
| **Software Engineering** | Git/GitHub, GitHub Actions (CI/CD), Python packaging, code review, public release and long-term maintenance of community-used code, Docker |
| **Machine Learning** | Autoencoders, physics-informed neural networks (PINNs), surrogate and reduced-order modelling, supervised and unsupervised methods |
| **Languages** | Italian (native), English (C2 – fluent), German (B1 – intermediate) |

## Work Experience

**Ludwig Maximilian University (LMU) Munich – Universitäts-Sternwarte** *Munich, Germany*

Research Scientist (Post-doctoral Researcher) *Apr. 2016 – Present*

- **Lead developer of the group's simulation framework.** Built and maintain a coupled radiation-hydrodynamics chain in C/Fortran/Python linking photoionisation/dust radiative transfer (MOCASSIN) and thermochemistry (PRIZMO) to the PLUTO MHD solver: governing equations, discretisation, solver coupling, validation, public release and maintenance. The model series built on it has 300 + citations. Co-author of the FARGOCPT code paper (A&A 684, A192) and of Hall-MHD disc-wind simulations (MNRAS 530, 5131).
- **Surrogate models for expensive physics.** Co-developed interpretable autoencoder surrogates that replace a full chemical reaction network: 10× smaller network, 2× shorter simulation wall-clock, accuracy checked against the full network on held-out data (Grassi et al. 2022, A&A 668, A139).
- **HPC campaigns.** >100,000 CPU-hours per year of peer-reviewed allocation since 2016 on national supercomputers (SuperMUC-NG and c2pap, LRZ Garching; ForHLR, KIT Karlsruhe): MPI/OpenMP, SLURM, profiling and scaling tests, automated post-processing. The public model grids are used directly as input physics by external disc-evolution and planet-population-synthesis codes.
- **Independent validation.** Co-authored the benchmark against a fuller thermochemical treatment that revised our own published mass-loss rates downward (Sellek et al. 2024, A&A 690, A296).
- **Research software practice.** Developing a department-wide framework for using LLM-based tools in research, with validation checkpoints and human review before any result is used; use LLM coding assistants for refactoring, test generation and documentation.
- **Supervision and teaching.** Day-to-day supervision of 4 doctoral and 6 master's students: defining projects and teaching the numerical methods and the codebase. Supervised students have led 16 refereed papers so far, including first-author papers from three master's theses, with a fourth in preparation. Collaboration with ESO, Harvard CfA and five European universities; taught scientific programming and planetary science (BSc/MSc).

**University of Tübingen – Institute for Astronomy & Astrophysics (IAAT)** *Tübingen, Germany*

Research Scientist (Post-doctoral Researcher) *Jun. 2014 – Mar. 2016*

- Group of Prof. W. Kley. Ran 3D SPH and grid-based fluid simulations (PHANTOM, FARGO, PLUTO) of coupled dust–gas dynamics; built Python pipelines for automated post-processing, statistical analysis and visualisation across HPC clusters.
- First-author modelling of the HL Tau system, among the first theoretical interpretations of the ALMA long-baseline images (A&A 584, A110).

## Education

**CISAS – Università degli Studi di Padova** *Padova, Italy*

PhD in Science, Technologies and Space Measurements (Curriculum: Astronautics and Satellite-based Science) *Jan. 2011 – Apr. 2014*

Numerical (SPH) modelling of circumstellar disc dynamics in binary systems; statistical analysis of the candidate target field for the ESA PLATO mission. Supervisors: Prof. F. Marzari, Dr. S. Desidera (INAF Padova).

**Università degli Studi di Padova** *Padova, Italy*

MSc in Astronomy (Laurea Specialistica) – 110/110 summa cum laude *Jan. 2007 – Sep. 2010*

**Università degli Studi di Padova** *Padova, Italy*

BSc in Astronomy (Laurea Triennale) – 103/110 *Sep. 2003 – Dec. 2006*

## Certifications & Training

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| 2022 | **SciML GPU Bootcamp**, Physics-Informed Neural Networks (PINNs), NVIDIA / Excellence Cluster ORIGINS Munich | *Online* |
| 2022 | **ORIGINS Data Science Block Courses**, Excellence Cluster ORIGINS, LMU Munich | *Online* |
| 2020 | **Fundamentals of Deep Learning**, NVIDIA Deep Learning Institute | *Munich, Germany* |
| 2019 | **Hybrid Programming in HPC: MPI + OpenMP**, Leibniz Supercomputing Centre (LRZ) | *Munich, Germany* |

## Selected Publications

**The dispersal of protoplanetary discs I: a new generation of X-ray photoevaporation models – 160 + citations** *MNRAS 487, 691*

**Picogna, G.**; Ercolano, B.; Owen, J. E.; Weber, M. L. *Jul. 2019*

**Reducing the complexity of chemical networks via interpretable autoencoders – machine learning applied to astrochemical modelling** *A&A 668, A139*

Grassi, T.; Nauman, F.; Ramsey, J. P.; Bovino, S.; **Picogna, G.**; Ercolano, B. *Dec. 2022*

**FARGOCPT: 2D Multiphysics code for simulating disk interactions with stars, planets, and particles** *A&A 684, A192*

Rometsch, T.; Jordan, L. M.; Moldenhauer, T. W.; Wehner, D.; Rendon Restrepo, S.; Müller, T. W. A.; **Picogna, G.**; Kley, W.; Dullemond, C. P. *Apr. 2024*

**Photoevaporation of protoplanetary discs with PLUTO+PRIZMO: I. Lower X-ray-driven mass-loss rates due to enhanced cooling** *A&A 690, A296*

Sellek, A. D.; Grassi, T.; **Picogna, G.**; Rab, C.; Clarke, C. J.; Ercolano, B. *Oct. 2024*

47 peer-reviewed publications (9 first-author), 1,300 + citations, h-index 22 (NASA ADS). Full list at NASA ADS and on my personal website.